Chi=test diagnostic

David M. Gregory, M.A.

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```
library(haven)
  library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
  library(tidyr)
library(RVAideMemoire)
```

```
## Warning: package 'RVAideMemoire' was built under R version 4.5.3

## *** Package RVAideMemoire v 0.9-83-12 ***
```

```
d <- read_dta("CMPS 2020_replication data.dta")
```

```
wtd_tab <- function(data, var, wt = "os_weight")
   {data %>% filter(!is.na(.data[[var]])) %>%
   group_by(value = as_factor(.data[[var]])) %>%
   summarise(n = n(), wtd_n = sum(.data[[wt]],
   na.rm = TRUE), .groups = "drop" ) %>%mutate(wtd_pct =
   round(100 * wtd_n / sum(wtd_n), 1))}
```

```
t1_vars <- c("S2_Racer5", "Q803", "S2_Race_Prime", "S2_Nativer2", "S2_Nativer3", "S2_Nativer4", "S2_Nat
```

```
subs <- list("Screener" = d%>%
filter(S2_Racer5 == 1), "Primary" = d%>%filter(S2_Race_Prime ==5),
"Census" =d%>% filter(Q803 == 3))
```

```r
sapply(subs, nrow)
```

```
## Screener  Primary   Census
##     1070      713      656
```

```r
t2_vars <- c("S2_M", "S6", "Q14",
"Q77r1", "Q21", "Q265", "Q57",
"Q266", "Q271", "Q819")
t2_vars <- c("S2_M", "S6", "Q14",
             "Q77r1",
             "Q21", "Q265",
             "Q57", "Q266",
             "Q271", "Q819")
```

```r
lapply(subs, function(x) table(x$Q803))
```

```
## $Screener
##
##   1   2   3   4   5   6   7
## 187 104 474  20  18 128 139
##
## $Primary
##
##   1   2   3   4   5   6   7
##  71  10 439  13  14  73  93
##
## $Census
##
##   3
## 656
```

```r
agree <- table(Screener = d$S2_Racer5 == 1, Census  = d$Q803 == 3, useNA = "ifany")

agree
```

```
##         Census
## Screener FALSE  TRUE
##    FALSE 16304   182
##    TRUE    596   474
```

```r
d_long <- d %>%
transmute(id      = row_number(),
screener = as.integer(S2_Racer5 == 1),
primary  = as.integer(S2_Race_Prime == 5),
census = as.integer(Q803 == 3)
) %>%
pivot_longer(-id, names_to = "method", values_to = "identified")

d_long <- d_long %>%
  mutate(method = factor(method, levels = c("screener", "primary", "census")))

cochran.qtest(identified ~ method | id, data = d_long)
```

```
##
##  Cochran's Q test
##
## data:  identified by method, block = id
## Q = 371.5793, df = 2, p-value < 2.2e-16
## alternative hypothesis: true difference in probabilities is not equal to 0
## sample estimates:
## proba in group screener  proba in group primary   proba in group census
##             0.06094782              0.04061290              0.03736614
##
##         Pairwise comparisons using Wilcoxon sign test
##
##           screener primary
## primary 2.044e-107       -
## census   5.808e-52 0.01142
##
## P value adjustment method: fdr
```